

Question: Hosmer-Lemeshow Goodness of Fit Test

A bank has fitted a logistic regression model to predict loan default using 200 historical loan applications.

The predicted probabilities have been sorted and divided into **10 groups** (deciles). The observed and expected counts for defaulters ($Y = 1$) and non-defaulters ($Y = 0$) in each group are shown below.

Answer the following:

- (a) State the null and alternative hypotheses for the Hosmer-Lemeshow test.
- (b) Compute the Hosmer-Lemeshow chi-square statistic χ_{HL}^2 .
- (c) Determine the degrees of freedom.
- (d) Using $\alpha = 0.05$, find the critical value and state your conclusion.
- (e) Interpret the result in plain language.

Group	Predicted Prob Range	Observed Defaults O_{1j}	Expected Defaults E_{1j}	Observed Non-Defaults O_{0j}	Expected Non-Defaults E_{0j}
1	0.00–0.05	0	0.5	20	19.5
2	0.05–0.10	1	1.4	19	18.6
3	0.10–0.20	2	3.0	18	17.0
4	0.20–0.30	4	5.0	16	15.0
5	0.30–0.40	7	7.0	13	13.0
6	0.40–0.50	10	9.0	10	11.0
7	0.50–0.60	12	11.0	8	9.0
8	0.60–0.70	14	13.0	6	7.0
9	0.70–0.80	16	15.0	4	5.0
10	0.80–1.00	18	19.1	2	0.9